engineering 4 us all

S Student Handout

Name: ______________________________ **Date:** _________________ **Class:** _______________

MyDesign Portfolio

Element I Initial Template

How to use this template

You must test *all* high priority requirements to score a 5. After testing, you will connect the data you collected to the "success" or "failures" of your design. You will consider next steps for your design based on the data you just analyzed. Each test that was listed in your table for Element H (testing plan) was on its own row. Generate a report for each test, the results, and its analysis that includes the following:

a) number and name of the test (as referenced in Element H)
    i) 1 - float test
    ii) 2 - collection test
b) indication of whether the test is high, medium, or low priority based on your requirements (as reference in Element C)
    i) Both are high priority tests that would make or break the ROV
c) for physical tests, include an explanation of your process and the full set of raw data such that the following best practices for scientific testing and mathematical modeling will be evident:
    i) Our process for the float test includes initial test in a bin filled with water, then going to the JCC and experimenting with different size floaties, and finally testing it in the ocean.
    ii) Our other test was the collection; we tested it in the ocean and gave it to mary.
d) summary of your data with appropriate charts, graphics, spreadsheets, etc., drawing on best practices in science and math for data display and analysis
    i) We had none
e) statement of what the data means and a summary of the implications of the data you collected
    - Based on our tests, we concluded that is does float and collect microplastics.
f) discussions about *each* implication from the summary of your data
    i) The implications of our data tell us that we finished and our ROV works.

Name: ______________________________ **Date:** _________________ **Class:** _______________

g) next steps for your design and process from the new information from the test that was conducted

i) No, it worked, and we are done.

Test 1. Must float

- High priority
- We put it in the pool at the JCC and we tested its ability to float. We adjusted the pool noodles on its sides to make it more balanced.

Test 2. Must collect microplastics.

High priority

It worked in the ocean and didn't break.